

Yu Qing (Perrie) Quek

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Education

University of Pennsylvania, School of Engineering and Applied Science

Philadelphia, PA

Master of Computer and Information Technology in Computer Science

Aug 2023 – Present

- **GPA:** 4.0/4.0
- **Relevant coursework:** Software Systems, Interactive Computer Graphics, Computer Systems Programming, Database and Information systems, Big Data Analytics, Data Structures and Algorithms, Software Development, Discrete Mathematics

Nanyang Technological University, School of Art, Design and Media

Singapore

Bachelor of Fine Arts in Media Art, Honours (Highest Distinction)

Aug 2017 – Jun 2021

- **Winner of the Lee Kuan Yew Gold Medal, 2020/2021:** Graduated top of class in a cohort of 160 students
- **GPA:** 4.79/5.00
- **Dean's list:** Awarded in academic Years 2017/2018, 2018/2019, 2020/2021
- **Scholarship:** Recipient of the NTU-University Scholars Programme Scholarship

Skills

Languages & Frameworks C++, Java, OpenGL, C, Python, C#, PostgreSQL, SQL, HTML/CSS, JavaScript, React.js, Node.js, Express, Vite

Software / Tools Docker, Git, Pandas, Adobe Creative Suite, QT Creator, Autodesk Maya, Dragonframe

Work Experience

Teaching Assistant

Philadelphia, PA

University of Pennsylvania

Aug 2024 – Present

- Teaching Assistant for CIT 5950 Computer Systems Programming and CIS 2400 Introduction to Computer Systems
- Provided code reviews in C and C++, hosted weekly office hours, maintained assignment infrastructure

Graduate Research Assistant

Philadelphia, PA

Autonomous Manufacturing Lab @ UPenn

Jun 2024 – Aug 2024

- Converted a Java-based codebase to C to integrate with Rhino3D (3D modeling software), optimizing workflow for architectural simulations
- Developed custom scripting components, enabling seamless interaction between simulation software and architectural models

Selected Projects

Distributed Multicast Chat Server (C++)

Software Engineer

Mar 2025

- Developed a fully distributed, UDP-based chat server supporting unordered, FIFO, causal order, and total order multicast across replicated servers without a central coordinator
- Implemented per-chatroom message broadcasting, vector clocks, and holdback queues to ensure consistent delivery and ordering

Multithreaded SMTP & POP3 Email Server (C++)

Software Engineer

Feb 2025

- Designed and implemented a multithreaded SMTP/POP3 email server from scratch, adhering to RFC standards
- Supported concurrent client connections with synchronization mechanisms for local and remote clients through Thunderbird

Mini Minecraft [\[link\]](#)

Graphics Programmer

Nov 2024 - Dec 2024

- Developed a mini version of Minecraft using C++ and OpenGL in a team of 3
- Implemented procedural terrain generation, day/night cycles, texturing, procedural assets, post-process shaders and more

Chartify | Music Trend Analytics Tool [\[link\]](#)

Full-Stack Developer, Data Engineer

Sep 2024 - Dec 2024

- Built a full-stack web application delivering music trend analytics through interactive data visualizations
- Designed and implemented a PostgreSQL database, integrating datasets from Spotify, Kaggle, and lyrics retrieved via LRCLIB API
- Developed and optimized SQL queries for efficient data retrieval